

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, AMT)

November 2018

ME784/ME734 Computer Aided Engineering

Date: 29.11.2018, Thursday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

- Q1 (a)** Using a simplified description of process of design explain where computational aids might be assistance to the designer. 5
- (b)** Answer following with correct option. 6
1. C^1 continuity refers to:
(A) Common tangent (C) Common curvature
(B) Common point (D) Common normal
 2. The degree of the Bezier curve with n control points is:
(A) $n + 1$ (C) $n - 1$
(B) n (D) $2n$
 3. Which attributes of image transformation rotate the image by a given angle?
(A) Rotate - X (C) Both (A) & (B)
(B) Rotate - Y (D) None of these
 4. Bresanham circle algorithm uses the approach:
(A) Midpoint (C) Point
(B) Line (D) None of these
 5. Types of computer graphics are:
(A) Vector and raster (C) Vector and scalar
(B) Scalar and raster (D) None of these
 6. In a display specified as 600 x 400 the number of pixels across the display screen is:
(A) 600 (C) 400
(B) 240000 (D) Both (A) & (B)
- Q2 (a)** Differentiate between World Coordinate System (WCS) and User Coordinate System (WCS). 3
- (b)** Explain the working principle of plotter. 3
- (c)** Using Bresenham's line algorithm, find the pixel positions between end points (2, 7) and (15, 10). 6

OR

- Q2 (a)** Explain the types of coordinate systems used in geometric modeling. 3
- (b)** State difference between raster scan and vector scan display. 3
- (c)** Explain generation of circle using Bresenham's Midpoint algorithm. 6

- Q3 (a)** Compare geometric transformation with coordinate transformation. Also explain composite transformation. **3**
- (b)** What are the advantages of homogenous coordinate system? **3**
- (c)** Perform 45 degree CW rotation of a triangle having coordinates A (0, 0), B (3, 2) and C (2, 3) about origin. Find the new vertex positions of triangle. **6**

OR

- Q3 (a)** What is concatenated transformation? Explain its benefits. **3**
- (b)** Explain 3D rotation transformations. Also give transformation matrix for each. **3**
- (c)** A triangle with vertices A (5, 5), B (8, 5) and C (5, 10) is to be reflected about the line $y = 3x + 4$. Determine the composite transformation matrix. **6**

SECTION- II

- Q4 (a)** What is reverse engineering? Explain Active Scanners. **4**
- (b)** Explain rapid prototyping. Explain FDM process. **3**

- Q5** One dimensional deformation inside the one meter long cast steel rod is governed by following equation. Find a three parameter solution using Galerkin Method. **7**

$$\frac{d^2u}{dx^2} + 10x = 0 \text{ in the range } 0 \leq x \leq 1$$

Subjected to $u(0) = 2$; $u(1) = 10$

OR

- Q5** Considering linear element, derive weak form and finite element formulation for the following one dimensional deformation governing equation. **7**

$$AE \frac{d^2u}{dx^2} = 0$$

- Q6** Attempt any *three*: **21**

- (a)** A wall of the length 100 mm is divided in to four equal elements using five nodes. Considering 1D heat transfer, steady state temperatures at node 1, 2, 3, 4 and 5 are 300, 250, 180, 150 and 100°C, respectively. Node 1 indicates left end of the rod and node 5 indicates right end of the rod. Find the temperature at point 30 mm and 90 mm from left end of the wall considering linear element.

- (b)** Discuss the steps involved in solution algorithm of the Gauss Seidel method for solving system of simultaneous equations.

- (c)** Using Tridiagonal matrix method, solve the following problem:

$$\begin{bmatrix} 5 & -2 & 0 & 0 \\ -1 & 5 & -1 & 0 \\ 0 & -2 & 5 & -2 \\ 0 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- (d)** Derive the finite difference formula to compute $d^2\phi/dx^2$ using central difference scheme
- (e)** Discuss applications of finite difference and finite element methods in solving engineering problems.
